

Curriculum Vitae

Employment

2025–present **Postdoc**, *Max Planck Institute for Mathematics in the Sciences (MPI-MiS), Leipzig.*

Education

2022–2025 **Ph.D. in Mathematics**, *IMB, Université de Bordeaux.*

2020–2022 **Master in Mathematics**, *Indiana University Bloomington.*

2017–2020 **Master in Mathematics**, *Nankai University.*

2013–2017 **Bachelor in Mathematics and Applied Mathematics**, *Beijing Institute of Technology.*

Research Interests

Keywords: Flat cone surfaces, translation surfaces, saddle connections, Delaunay triangulations, moduli spaces, linear subvarieties.

My research centers on surfaces (real two-dimensional manifolds), with an emphasis on the connections between geometry, combinatorics, and dynamics. In particular, I study the geometry and dynamics of translation surfaces and flat cone surfaces, as well as structures on their moduli spaces.

Publications and Preprints

Siegel–Veech Measures of Convex Flat Cone Spheres, [arXiv:2504.14731](https://arxiv.org/abs/2504.14731), to appear in *Compositio Mathematica*, 2026.

Uniform Length Estimates for Trajectories on Flat Cone Surfaces, [arXiv:2409.14188](https://arxiv.org/abs/2409.14188), to appear in *Annales de l'Institut Fourier*, 2026.

Bounds on saddle connections for flat spheres, with Guillaume Tahar, [arXiv:2308.08940](https://arxiv.org/abs/2308.08940), to appear in *Journal of Topology*, 2026.

In Preparation

L^∞ -Delaunay cells and linear subvarieties in the moduli space of Abelian differentials, with Vincent Delecroix and Bradley Zykoski.

Programming Project

I am interested in **SageMath**, an open-source mathematics software system built on Python. Currently, I am working with Vincent Delecroix on the computation of real linear subvarieties in the moduli spaces of Abelian differentials. More precisely, we develop algorithms to determine whether a given linear subspace in a period chart is tangent to a linear subvariety. We also compute the boundary of a linear subvariety in the multiscale compactification. This project is being implemented in the package **veerer**.

Talks

- Nov 25, 2025 **The floor to the junior researchers of Plato's tetrahedron's research network**, *Université de Montpellier*, Montpellier, France.
- Nov 18, 2025 **UCL/KCL Junior Geometry Seminar**, *King's College*, London, United Kingdom.
- Oct 08, 2025 **Géométrie et dynamique dans les espaces de modules**, *Institut Henri Poincaré (IHP)*, Paris, France.
- Sep 25, 2025 **Fifth GALS meeting**, *Institut Henri Poincaré (IHP)*, Paris, France.
- Aug 25, 2025 **Seminar**, *School of Mathematics, Sun Yat-sen University*, Guangzhou, China.
- Aug 15, 2025 **Dynamical Systems Seminar**, *Shanghai Institute for Mathematics and Interdisciplinary Sciences*, Shanghai, China.
- Jul 29, 2025 **Seminar**, *Chern Institute of Mathematics*, Tianjin, China.
- Jul 17, 2025 **The 5th Korea-France Conference in Mathematics**, *Institut de Mathématique de Bordeaux*, Bordeaux, France.
- Jun 27, 2025 **Séminaire de Géométrie**, *Institut de Mathématique de Bordeaux*, Bordeaux, France.
- Nov 08, 2024 **Séminaire de Systèmes Dynamiques**, *Institut de mathématiques de Toulouse*, Toulouse, France.
- Nov 07, 2024 **Groupe de travail**, *Institut de mathématiques de Toulouse*, Toulouse, France.
- Oct 04, 2024 **Geometry Topology Grad and Postdoc Seminar**, *University of Toronto*, online.
- Sep 10, 2024 **Riemann Surfaces: Random, Flat, and Hyperbolic Geometry**, *MFO*, (Lightning talk), Oberwolfach, Germany.
- Oct 17, 2023 **Cours de Géométrie Algébrique et Géométrie Complexe de la fédération MARGAUX**, *Le Laboratoire de Mathématiques et Applications, Université de Poitiers Poitiers*, (Short talk), Poitiers, France.
- Jun 08, 2023 **Student Seminar**, *YMSC, Tsinghua University*, Beijing, China.
- Mar 16, 2023 **Reading group on mapping class groups**, *Institut de mathématiques de Bordeaux*, Bordeaux, France.

Visiting

- Aug, 2025, School of Mathematics, Sun Yat-sen University (1 week)
- Jul, 2025, Chern Institute of Mathematics (1 week)
- Jun, 2023, YMSC, Tsinghua University (3 weeks)

Organization

- Mar, 2024 Co-organization of the School on flat surfaces and interactions, Le Teich
Jan - May, 2023 Co-organization of the Reading group on mapping class groups, Bordeaux

Conference Participation

- Sep, 2025 Fifth GALS meeting, IHP, Paris
Jul, 2025 The 5th Korea-France Conference in Mathematics, IMB, Bordeaux
Nov, 2024 Riemann Surfaces: Random, Flat, and Hyperbolic Geometry, MFO, Oberwolfach
Jun, 2024 New Frontiers in Parabolic Dynamics and Renormalization, Department of Mathematics, University of Bologna, Bologna
Jun, 2024 Rencontre GALS, Institut Camille Jordan, Lyon
Mar, 2024 School on flat surfaces and interactions, Le Teich
Jan, 2024 Sage Days 125: combinatorics, number theory, dynamical systems and geometry, Le Teich
Jan, 2024 Research School - Hodge Theory and o-minimality, CIRM, Marseille
Dec, 2023 Renormalization and Visualization in Geometry, Dynamics and Number Theory, CIRM, Marseille
Nov, 2023 The floor to the junior researcher of Plato's network Institut Fourier, Grenoble
Oct, 2023 Cours de Géométrie Algébrique et Géométrie Complexe de la fédération MARGAUX, Université de Poitiers, Poitiers
Oct, 2023 Probability and Geometry in, on and of non-Euclidian spaces, CIRM, Marseille
Jul, 2023 Renormalization and Visualization for packing, billiard and surfaces, CIRM, Marseille
Feb, 2023 Sage Days 117: combinatorics, number theory, dynamical systems and geometry, Le Teich
Sep, 2022 Combinatorics, Dynamics and Geometry on Moduli Spaces, CIRM, Marseille
May, 2022 Structures on Surfaces, CIRM, Marseille
Apr, 2022 Mini-rencontre ANR MoDiff, LaBRI, Université de Bordeaux, Bordeaux
Jun, 2021 CMI-HIMR Dynamics and Geometry Online Summer School, University of Bristol

Languages

- Chinese native
English fluent
French basic